wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Disconnect the headlamp and dash wire harness connector for the brake warning indicator switch from the switch terminals. Check for continuity between the red brake warning indicator driver circuit (G99) cavity of the headlamp and dash wire harness connector for the brake warning indicator switch and a good ground. There should be no continuity. If OK, go to Step 2. If not OK, repair the shorted red brake warning indicator driver circuit (G99) between the brake warning indicator switch and the instrument cluster as required.

(2) Disconnect the floor wire harness connector for the park brake switch from the switch terminal. Check for continuity between the red brake warning indicator driver circuit (G9) cavity of the headlamp and dash wire harness connector for the brake warning indicator switch and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted red brake warning indicator driver circuit (G9) between the brake warning indicator switch and the park brake switch as required.

(3) Check for continuity between each of the two terminals of the brake warning indicator switch and a good ground. In each case, there should be no continuity. If OK, go to Step 4. If not OK, replace the faulty brake warning indicator switch.

(4) Check for continuity between the terminal of the park brake switch and a good ground. There should be no continuity with the park brake released, and continuity with the park brake applied. If not OK, replace the faulty park brake switch.

# CHECK GAUGES INDICATOR

## DESCRIPTION

CHECK
GAUGES

***Fig. 10 Check Gauges Indicator***

A check gauges indicator is standard equipment on all instrument clusters (Fig. 10). The check gauges indicator is located on the lower edge of the instrument cluster, to the right of center. The check gauges indicator consists of a stencil-like cutout of the words “CHECK GAUGES” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “CHECK GAUGES” text to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The check gauges indicator is serviced as a unit with the instrument cluster.

## OPERATION

The check gauges indicator gives an indication to the vehicle operator when certain instrument cluster gauge readings reflect a condition requiring immediate attention. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The check gauges indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the check gauges indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the check gauges indicator is illuminated for about two seconds as a bulb test.
- **Engine Temperature High Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is about 127° C (261° F) or higher, the check gauges indicator will be illuminated. The indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine temperature is about 124° C (255° F) or lower, or until the ignition switch is turned to the Off position, whichever occurs first.
- **Engine Oil Pressure Low Message** - Each time the cluster receives a message from the PCM indicating the engine oil pressure of a 2.4L engine is below about 0.2 $kg/cm^2$ (3 psi), or of any engine other than a 2.4L is below about 0.4 $kg/cm^2$ (6 psi), the check gauges indicator will be illuminated. The indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine oil pressure of a 2.4L engine is above about 0.2 $kg/cm^2$ (3 psi), or of any engine other than a 2.4L is above about 0.4 $kg/cm^2$ (6 psi), or until the ignition switch is turned to the Off position, whichever occurs first. The cluster will only turn the indicator on in response to an engine oil pressure low message if the engine speed is 300 rpm or greater for more than about five seconds.
- **System Voltage Low (Charge Fail) Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is about 9.0 volts or lower (charge fail condition), the check gauges indicator will be illuminated. The indicator